Three-board GPIO daisy chain

Autonomous sensor, relay/dimmer and lighting control with live OLED diagnostics

Sense locally.
Move packets.
Show the system working.

NuMaker NDA102SD2 boards linked through DA_JP1 TX/RX test points, with a digital TSL25721 ambient sensor.

01 SENSOR
PB5 + digital ALS

02 RELAY
Motion + ambient PWM

03 LIGHTING
RGB + warm/cool

3
autonomous nodes

1000 baud
software UART

15 s / 10 s
warmup / motion burst

Bench firmware and verification artifacts

Three local applications form one visible system

Each node owns its hardware; addressed packets carry readings and output state.

01 SENSOR / ORIGIN

PB5 voltage via MCU ADC CH5

TSL25721 over I²C

Sample locally; publish once per second

02 RELAY / ROUTER

PC2 active-low motion relay

PA6 ambient dimmer PWM

Authoritative 10-second event timer

03 LIGHTING / ACK

Red/blue motion strobe

Smooth warm/cool white fade

Return state and acknowledgment

SENSOR → RELAY → LIGHTING → SENSOR | USB supplies optional diagnostics

Bench firmware and verification artifacts

Wire TX to the next board’s RX

DA_JP1 logic-side test points • common ground • compatible logic supplies

FROM	TO	PURPOSE
Sensor D_TX / PC14	Relay D_RX / PB15	Sensor readings + motion requests
Relay D_TX / PC14	Lighting D_RX / PB15	Relay state + dimmer + event timer
Lighting D_TX / PC14	Sensor D_RX / PB15	State + motion acknowledgment
Sensor GND	Relay GND + lighting GND	Shared signal reference

Keep one transmitter per receive wire; the third board needs the return link to close the ring.

Power each board correctly. Do not connect the DALI bus-voltage terminals to these logic jumpers.

Bench firmware and verification artifacts

GPIO software UART on DALI test points

The header location and the waveform protocol are separate design choices.

IMPLEMENTED ON THE JUMPERS

1000 baud, 8 data bits, no parity, 1 stop bit

PC14 open-drain TX; PB15 input

20 kHz SysTick services TX/RX queues

DALI INTERFACE CONTEXT

The baseboard includes a DALI transceiver interface with bus conditioning and isolation circuitry.

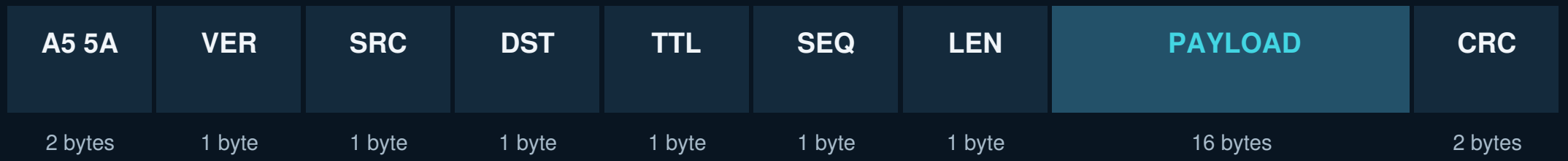
This demo exchanges custom serial frames at the MCU-side test points.

The current demo uses a private UART-style packet protocol; it does not implement DALI framing.

Bench firmware and verification artifacts

A compact frame carries addresses, state and integrity

CRC16 rejects damaged frames; TTL bounds generic forwarding.



TRANSPORT RULES

1–16 payload bytes; 26 bytes for the demo
CRC16-CCITT-FALSE, big-endian CRC
Destination consumes; others decrement TTL

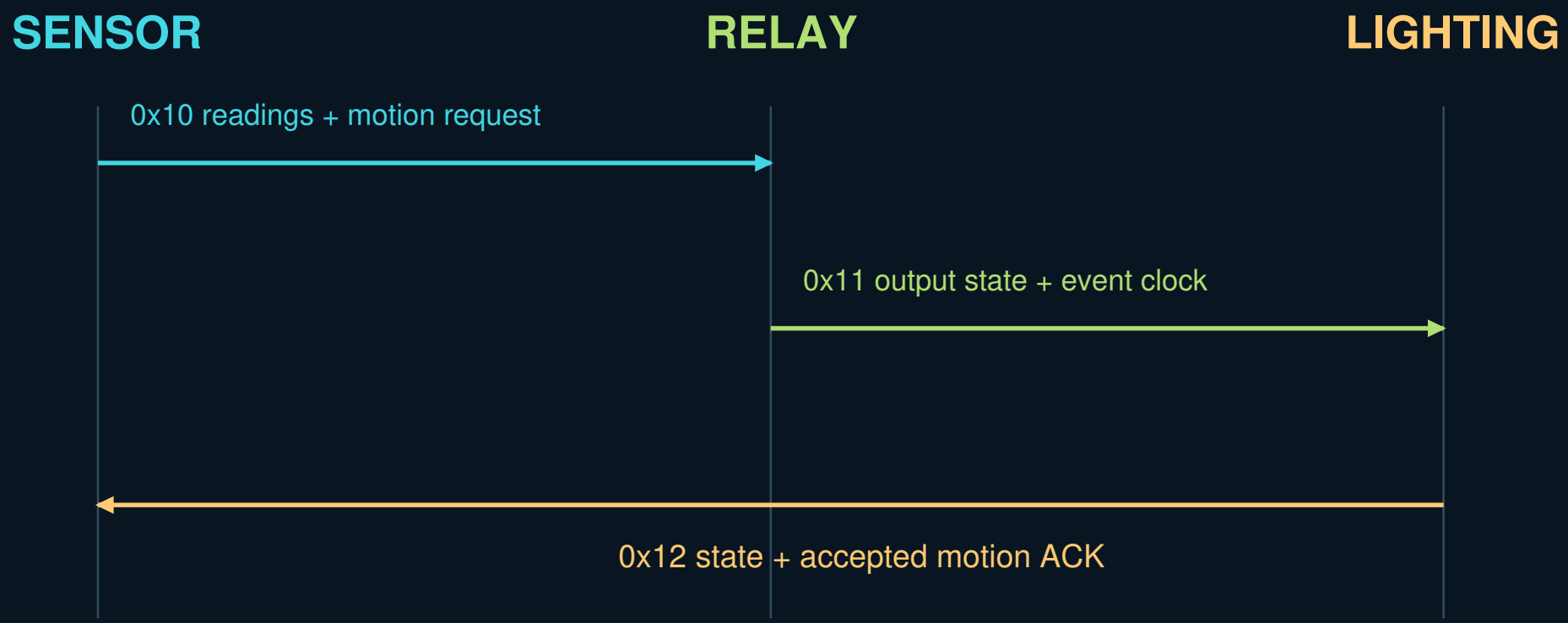
DEMO PAYLOAD / VERSION 2

Validity flags, PB5 raw count, lux
PWM %, relay, event ID + elapsed ms
Warmup, ready mask, motion ID + ACK

Bench firmware and verification artifacts

One reading becomes an acknowledged output state

The source sends at 1 Hz; the two downstream nodes respond to received packets.



26 bytes × 10 serial bits ÷ 1000 baud = 260 ms airtime per hop; about 780 ms around the ring.

Bench firmware and verification artifacts

15-second warmup, then a fixed 10-second burst

Warmup protects startup; readiness and PIR rearming handle runtime conditions.

1 WARMUP

Each MCU counts 15 s.
The ring and ambient paths keep running.

2 READY + CLEAR

Fresh S/R/L bits reach 111.
PB5 is valid and has been inactive long enough to arm.

3 FIRST TRIGGER

Accept a fresh motion ID.
Run relay + strobe for 10 s; repeated HIGH does not extend it.

A new burst requires PB5 to clear and rise again. A held HIGH is not a new motion edge.

Bench firmware and verification artifacts

Motion and ambient light have separate state

They share a CPU and ring transport, while keeping trigger and brightness decisions independent.

PB5 / MOTION PATH

MCU ADC CH5 → raw PB5
50 ms target sample interval

Hysteresis → debounce → motion ID
Fixed relay/strobe deadline + ACK

TSL25721 / AMBIENT PATH

I²C registers → two raw channels → lux
200 ms target polling interval

Lux validity → PWM target → local ramp
Tunable-white brightness + color mix

Ambient read errors retain the last dimmer target; they do not reset motion capture or its deadline.

Bench firmware and verification artifacts

The TSL25721 performs its own analog conversion

The MCU reads digital I²C registers; its PB5 ADC is a separate measurement path.

CH0 + CH1

CH0: visible + infrared
CH1: primarily infrared

Two on-sensor integrating ADCs

DIGITAL DATA

Two 16-bit result registers

I²C slave address 0x39
Observed chip ID: 0x34

SENSOR MCU

PC0 = SDA; PC1 = SCL
Software I²C master

Compute lux and publish it
No MCU ADC for ambient light

Separate buses: ambient sensor on PC0/PC1 at 0x39; OLED on PB0/PB1 at 0x3C. Neither is the PC14/PB15 GPIO ring.

Sources: ams OSRAM TSL2572 DS000178 pp.14–15; bench identification; sensor firmware

Configure gain and integration, then enable ALS

Values below are command bytes used on the wire, including the register-command bit.

STEP	COMMAND / VALUE	WHAT IT DOES
Identify	Read 0x92 → expect 0x34	Verify the TSL25721 identity
Power	Write 0x80 = 0x01	Enable oscillator / PON
Wait	At least 3 ms, nonblocking	Allow startup before ALS configuration
Integration	Write 0x81 = 0xED	(256 – 237) × 2.73 ms = 51.87 ms
Gain + config	0x8F = 0x00; 0x8D = 0x00	1× gain; standard gain mode
Acquire	Write 0x80 = 0x03	PON + AEN; continuous ALS conversions

Sensor-register offsets are 0x12, 0x00, 0x01, 0x0F and 0x0D; 0x80 selects the command register.

Sources: sensor firmware; ams OSRAM DS000178 register map, ENABLE, ATIME, CONFIG and CONTROL

Read four bytes, then assemble two 16-bit counts

The data registers are digital values produced by the sensor’s internal ADCs.

```
START 0x72 W 0xB4 RESTART 0x73 R
C0_LO C0_HI C1_LO C1_HI NACK STOP
```

REGISTER AND BUS DETAILS

0x39 is the 7-bit slave address.
0x72 / 0x73 are address + R/W bytes.
0xB4 = 0xA0 | 0x14: auto-increment from channel-0 low byte.

ASSEMBLY + VALIDITY

C0 = byte[0] | byte[1] << 8
C1 = byte[2] | byte[3] << 8

Read 0x93 first; require AVALID bit 0.
Reject either channel ≥19456 counts.

Sources: sensor firmware demo_sensor_io/demo_lux_sample; ams OSRAM DS000178 pp.18–20, 27

Compensate infrared, then scale by integration and gain

Measured example: C0 = 751 and C1 = 122 → approximately 604.81 lux.

```
CPL = (51.87 ms × 1) / (1 × 60)
= 0.8645 counts / lux

L1 = (C0 − 1.87 × C1) / CPL
L2 = (0.63 × C0 − C1) / CPL
Lux = max(L1, L2, 0)
```

WORKED REGISTER READ

Bytes: EF 02 7A 00

C0: 0x02EF = 751
C1: 0x007A = 122

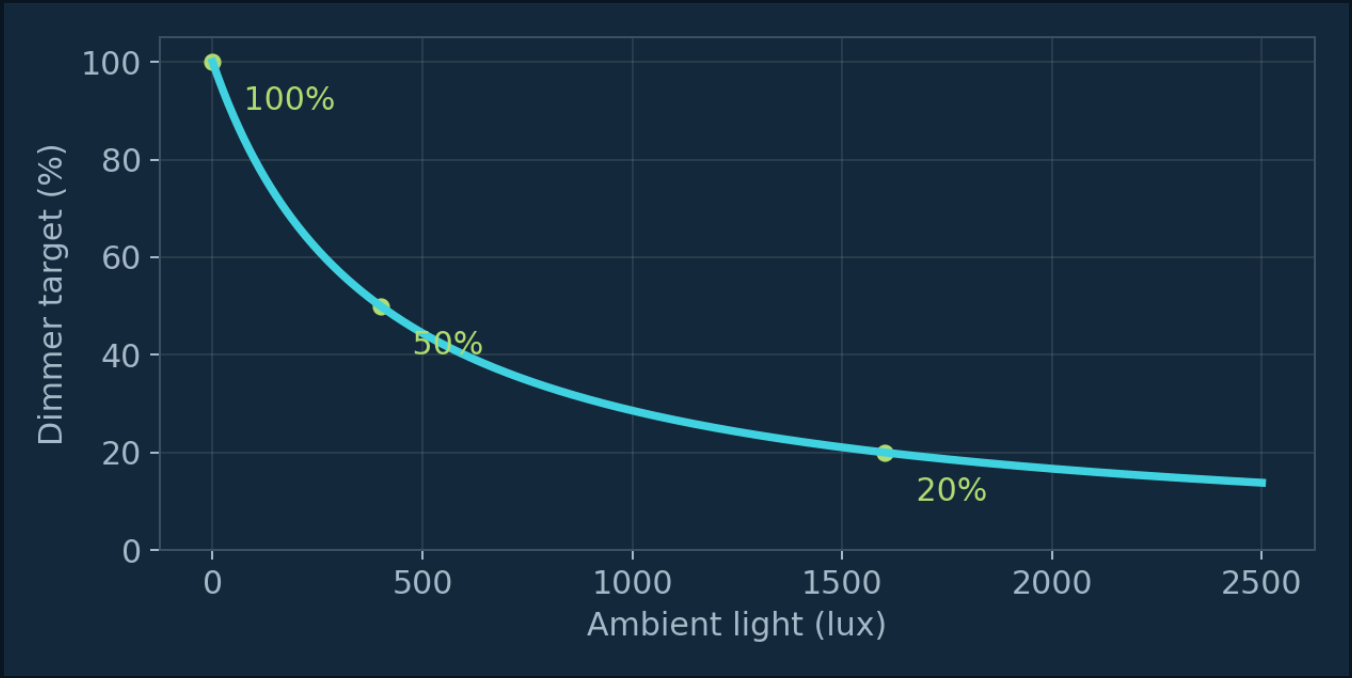
Float calculation: 604.81 lux
MCU integer result: 604 lux

The prototype uses glass factor 1 and fixed gain. Values are estimates; no enclosure or optical calibration has been applied.

Sources: ams OSRAM DS000178 Lux Equation; build/ambient-sensor-working.json; integer firmware implementation

Lux controls brightness; dimmer duty controls white mix

Motion drives relay/RGB. Ambient light independently drives PWM and warm/cool white.



CONTROL LAW

PWM target = round(40000 / (400 + lux))

0 lux → 100% | 400 lux → 50%
7000 lux → 5%

Relay ramps one point per 80 ms.
White fade: up to 12.5 points/s.

For smoothed duty b: warm = round(b^2 / 100), cool = b – warm. Example: b = 50 → 25 / 25.

Bench firmware and verification artifacts

PB5 is a pin; its displayed number is an ADC count

A two-level signal can be measured by an ADC without being an analog motion-strength output.

CURRENT FIRMWARE PATH

PB5 physical pin
↓ temporary analog pin function
MCU EADC channel 5
↓ 12-bit voltage measurement
Displayed counts: typically ~1 or ~1900+

WHAT WE KNOW / STILL NEED

Two observed count clusters suggest an on/off signal.

Exact PIR module output type and active polarity still need schematic or part identification.

Verify actual voltages before a GPIO-only read.

Current trigger thresholds: HIGH >1800, LOW <100. Intermediate counts retain the motion candidate.

Bench firmware and verification artifacts

A raw PB5 HIGH pulse lasted 19.6 seconds

The long input pulse outlives the fixed 10-second output interval.



OBSERVATION

19.611 s raw HIGH

Sampling and packet traffic continued.

Cover/uncover trials correlated with longer pulses.
Cause remains open.

After the 10-second burst, a held HIGH cannot create another rising edge; the input must clear first.

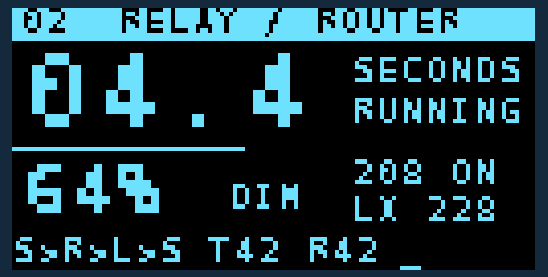
Source: actual PB5 log pir-live-20261111-202854.jsonl; measured completed HIGH = 19611 ms

Put live state where the wiring is

Normal dashboards visualize timing and packets; the sensor diagnostic enlarges the raw input.

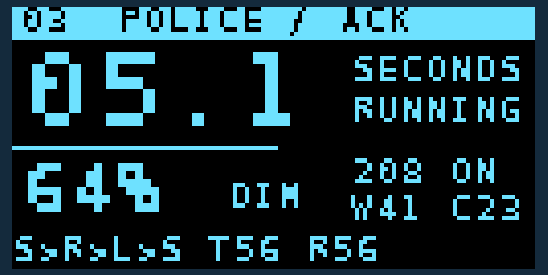


Large-PB5 diagnostic • native-rendered preview



Relay dashboard

Captured framebuffer



Lighting dashboard

Captured framebuffer

Sensor diagnostic: 28-pixel digits, current level duration, last HIGH duration, warmup/readiness and packet counts.

Bench firmware and verification artifacts

Separate verified behavior from open hardware questions

Bench evidence supports the software paths; unplugged-power behavior and PIR causality need distinct checks.

VERIFIED

Three baseline images flashed and verified

All-board ready mask reached 7
Traffic and both samplers advanced with host serial ports closed for 12 s

Native protocol, timer, rearm and sensor-failure tests pass

OPEN / LIMITED

Serial-ports-closed test kept USB cables attached; it does not prove physical unplugged-power operation.

PIR output type/polarity and long-pulse cause remain unconfirmed.

Ring has airtime and scheduling latency.

Verification artifacts retain timestamps, port/node IDs, counters, output state and flash results.

Bench firmware and verification artifacts

Compare ambient control with a fixed light output

Keep sensing and logging active while removing dimmer changes from the experiment.

A BASELINE

Normal ambient dimming

Record PIR-only pulses, then repeat the cover/uncover action.

Keep the hand away from PIR where possible.

B FIXED DUTY

Diagnostic relay build: 60%

Lux and PB5 keep updating. PWM and white target stay fixed.

Repeat the same action after the white fade settles.

COMPARE

Log HIGH duration, lux, duty, event ID and sample counters.

A difference points toward an output-related effect; more controls distinguish optical from supply effects.

Status: the fixed-60% relay image is built and native-tested; deployment is pending relay USB connection.

Bench firmware and verification artifacts

Internal MCU flash stores the application

All three targets are NDA102SD2. The build allocates 64 KB APROM and 12 KB RAM.

REPRODUCIBLE BUILD

python3 tools/build.py

Sensor: --profile sensor --standalone-demo
Relay: --profile dt57 --relay-active-level 0 --standalone-demo
Lighting: --profile lighting --standalone-demo --swap-green-blue

GUARDED FLASH + DIAGNOSTICS

Identify the exact target before erase.
Snapshot current APROM.
Erase → write → verify → reset.

USB commands are read-only in the autonomous application. External SPI flash is not used by this demo.

Lighting-board override: PC5 red, PC3 green, PC4 blue. The observed green/blue wiring differs from schematic labels.

Bench firmware and verification artifacts

Use the implementation and captured evidence together

External references establish device behavior; local artifacts document what this prototype actually did.

Nuvoton baseboard reference	NuMaker-DALI-NDA102SD2 User Manual — DA_JP1 / PC14 / PB15
Ambient sensor primary source	ams OSRAM TSL2572 DS000178 v2-00 — ADCs, registers, I²C and lux
Implementation	firmware/header_link.h, header_ring.h, standalone_demo.h, demo_startup.h, demo_effects.h
Bench evidence	build/standalone-startup-verification.json; ambient-sensor-working.json; pir-live-*.jsonl
Diagnostic status	build/pb5-display-diagnostic.json; fixed-dimmer-diagnostic.json
Official branding assets	Nuvoton and Winbond official website logo images

Bench firmware and verification artifacts